

Situation report, September 18, 2026

Where the model stands after the overnight tests, and what to decide

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Contents

The short version	1
1. The mechanism, with the evidence	2
2. Decisions, with their evidence	3
3. What ran overnight, what failed, what is still running	5
4. Baseline and provenance	6
5. Next steps, in order	6

The short version

The baseline is solid: the paper's code is on main, tagged, and the September 14 initial state replays exactly. The sandbox solves one change at a time in a few minutes on that code. Overnight it produced the first full set of one-change comparisons, and they change what the paper can claim.

1. **The mechanism on the slide does not operate at this calibration.** Removing the down payment raises ownership by ten points and leaves fertility and its timing unchanged; giving households unsecured credit raises fertility and brings births forward but leaves ownership unchanged. Making space scarce (rooms at the data's level) does not change this. At these parameters, births respond to liquidity and to the price of space, not to the housing-collateral requirement, which governs ownership.
2. **Children at home are net costs at the retained parameters.** The value of a household falls at every state when a child stays one more period. Fertility is held up by the taste shocks and by the child-preference level being set to hit 2.1. This is the P1 problem in its sharpest form, and it means any change that lengthens dependency or adds a child cost must be compared with that level re-normalized, not fixed.
3. **The maturation fix works.** A parent-age exit law with a newborn exemption makes children at home by parent age match the ACS at both ends, where the current law is 40 percent short at ages 30–42 and keeps adult “dependents” past 58.
4. **Two switches move the fit a lot, in the right places.** A size-dependent rental wedge in place of the cap and the owner premium cuts the loss from 2607 to 402 (ownership overshoots and would be

calibrated down). A 20 percent earnings cost of children at home cuts it to 1746 and moves first births later and the rooms response toward its target.

5. **Earnings:** one consistent PSID decomposition brings the old-age wealth tail to its target (3.38 against 3.52); the Boar–Gorea–Midrigan process without permanent types loses the tail (2.9) and gives too few childless households. Permanent types are what the tail buys.

1. The mechanism, with the evidence

All numbers below are steady states at the retained parameters. “Fixed ψ ” holds the child-preference level at its retained value; “ ψ root” re-sets it so completed fertility is 2.1. The sandbox’s baseline is close to, not equal to, the slides’ levels (its ownership is 0.46 against the slides’ 0.54; the reason is a step in the original launch script the sandbox reproduces only approximately). Differences between columns are the object; levels are not.

Frictionless benchmark (fixed ψ).

Moment	Baseline	No down payment ($\varphi = 1$)	Unsecured credit line (five years’ earnings)
Completed fertility	1.872	1.866	1.948
Childless 40–44	0.239	0.241	0.225
Mean first-birth age	26.96	26.97	25.79
Ownership 30–55	0.459	0.557	0.450
Wealth / earnings	5.18	5.07	4.43
First-birth rooms response	1.07	1.15	1.17

Who is constrained. The down payment binds for 4.5 percent of family-forming households (ages 26–38, zero or one child at home). With the supply scale lowered until mean rooms equal the data (5.56, price up 4 percent), that share falls to 2.8 percent, and fertility is flat with and without the down payment (1.840 against 1.835). Scarcer space lowers fertility a little through cost (1.872 to 1.840). The author’s prior that the rooms misses hide the mechanism is rejected on this test.

Why. A renter can have six rooms and the parents’ space floor is 2.3 rooms, so a family with one or two children fits in a rental; the owner-only sizes start at eight rooms and matter for three-plus families. What blocks an early birth is that a renter cannot borrow at all and must save the child’s cost in advance. That is a precautionary channel, and it is real, but it is not the collateral story.

What survives. Easier mortgages raise ownership and not births; easier unsecured credit raises births and not ownership; housing costs lower fertility through the price of space. The property-tax channel to young families runs through ownership and the allocation of large homes, so its fertility effect will be small unless the space margin binds. The slide’s mechanism sentence has to change; the model, the calibration machinery and the experiment stand.

2. Decisions, with their evidence

Every row of the table below has completed fertility 2.1 (ψ re-normalized). Baseline loss 2607, ψ 0.190. Losses use the sandbox's twelve-row objective and are comparable across rows, not with the slides' 179.

Change	Loss	ψ	Childless	First- birth age	Ownership 30–55	Old p90/ p50	Rooms resp.	Recent- parent gap
Baseline	2607	0.190	0.201	26.1	0.455	4.32	1.12	0.47
Rental wedge, no cap, $\chi = 1$	402	0.205	0.208	26.3	0.730	3.83	0.97	0.28
Earnings penalty 20 % while chil- dren home	1746	0.315	0.228	27.1	0.416	4.46	0.85	0.39
Mortgage block (origination- only, 11 % amorti- zation)	2050	0.189	0.201	26.1	0.349	5.05	1.08	0.42
Boar- Gorea- Midrigan earn- ings, no types	2100	0.156	0.144	27.0	0.499	2.92	1.24	0.41
E6b earn- ings, one de- compo- sition	2325	0.180	0.184	26.6	0.477	3.38	1.14	0.44
Tenure shock $\kappa_H = 0$	2379	0.190	0.202	26.1	0.451	4.30	1.12	0.45
Parent- age matu- ration	2464	0.167	0.210	26.5	0.449	4.36	1.39	0.45

Change	Loss	ψ	Childless	First- birth age	Ownership 30–55	Old p90/ p50	Rooms resp.	Recent- parent gap
Modest unse- cured credit line	2539	0.187	0.202	25.8	0.446	4.32	1.15	0.46
Concave child benefit (log)	2111	0.199	0.210	25.9	0.449	4.15	1.10	0.43
Estate trans- fer to ages 45–65	2582	0.181	0.193	26.4	0.446	4.07	1.06	0.46
All switches to- gether	962	0.480	0.223	27.6	0.426	4.44	0.97	0.27
Rental cap at 8 rooms (no other change)	489	0.189	0.202	26.1	0.300	4.38	1.36	0.23

Targets: childless 0.198, first-birth age 26.0, ownership 0.648, p90/p50 3.52, rooms response 0.72, recent-parent gap 0.16.

F3, how children leave home. Recommend the parent-age law with the newborn exemption (exit 0.05 per period for young parents, rising from parent age 34 to certain exit at 62). Evidence: with ψ re-normalized, the ACS profile matches at 22–34 and is zero after 62; the two rooms moments rise (a refit would lower the space floor). The switch is built, tested and verified line by line; off by default.

P1, the child term. Two facts: children at home are net costs at every state, and the intensive margin is carried by the taste scales. Candidates: keep linear and say so; a concave benefit; or the earnings penalty (S3), which at 20 percent halves the loss, moves first births later and the rooms response toward target, and needs ψ to double. The concave benefit (log) cuts the loss to 2111 with the first-birth age exactly on target (25.9 against 26.0) and fewer large families, at ψ 0.199. Recommend both: the concave benefit for the intensive margin and S3 with the penalty taken from the child-penalty literature for the income gradient, keeping ξ only if childlessness still needs it. Author decision.

H3/H6, the cap and the premium. The size-dependent wedge is the single largest improvement and it also disciplines the ownership level from the rent–price ratio rather than from a taste parameter. Its intercept is too strong at 0.02 (ownership 0.73); it would be calibrated. Recommend adopting the wedge and retiring χ ,

with the AHS renter share by size as its moment.

H1, the mortgage block. Amortization with origination-only borrowing lowers ownership by ten points and raises old-age wealth dispersion; it does nothing to fertility. It makes the model's finance side standard and the Coven comparison like for like. Recommend adopting it and letting the wedge intercept absorb the ownership level. A modest unsecured credit line brings first births forward by a third of a year and is the cheapest change that improves the timing rows; recommend it in the baseline at the Kaplan–Violante size.

E1, earnings. The permanent types are what the old-age wealth tail buys; without them (Boar–Gorea–Midrigan) the tail is 2.9 against 3.5 and childlessness falls to 0.14. The one-decomposition E6b process puts the tail at 3.38. Recommend retaining the types with the E6b decomposition, one tax treatment throughout.

F2, the tenure shock. Inert in the steady state; loss slightly better at zero. Recommend zero, with the kink handled numerically.

P5, estates. At fixed ψ the transfer to ages 45–65 barely moves anything (loss 2599) and costs an hour per solve; the ψ -root result is on Torch. Recommend the net-of-selling-cost valuation regardless; the receiver is a welfare-accounting choice more than a fit lever.

3. What ran overnight, what failed, what is still running

- Built and verified, all default off, full suite 227 tests: parent-age maturation (with an exact second housing-stage solve after my rejection of a shortcut); child earnings penalty; mortgage block; rental wedge; estate receiver. Muse sessions died silently three times on long tasks; the work was recovered from disk each time and every switch was re-run through the suite. Two sandbox bugs found and fixed on the way: a ϕ override silently reset by the package, and a spec loader without list support.
- Solved locally: the frictionless pair, the scarce-space pair, the who-is-constrained tables, the ACS profiles, eight fixed- ψ single switches, and eight ψ -root single switches (see the table).
- On Torch: ten ψ -root jobs (ids 17938995–17939004) and the smoke were queued from 01:30 in a congested queue (about 2,800 jobs pending) and at 11:00 all show CANCELLED with zero elapsed time: they never ran and were cancelled by a user or an administrator, not by this session. The batch folder on scratch was missing one folder the launcher expects; it was repaired at noon, the smoke passed in 39 seconds, and the two rows that depend on it (estate transfer and all switches with ψ re-normalized; each estate iteration re-solves the equilibrium, so they are too slow for the laptop) were resubmitted as jobs 17949060 and 17949061 and completed in 2.2 and 8.8 hours; their rows are in the table. Estate transfer: loss 2582 (2607), first births later by a third of a year, old-age tail 4.07 (4.32), nothing else moves. All switches together: loss 962, ψ 0.480 (two and a half times the baseline root), first-birth age 27.6 against 26.0, first births 30+ 0.32 against 0.25, wealth/earnings 4.4, rooms 5.25 (below the 5.56 target for the first time), three-plus rooms gap 0.58 (target 0.35). The switches do not add up: the penalty and the maturation law each need a higher ψ and together push timing past the target, while the wedge alone fits better (402) than all of them together.
- Added September 19: rental cap raised from six to eight rooms, nothing else changed. Fixed ψ : completed fertility 1.876 against 1.872, childless 0.238 against 0.239, first-birth age unchanged, ownership 0.31 against 0.46, recent-parent gap 0.21 against 0.46. ψ root: loss 489, ψ 0.189 (baseline 0.190). On the paper's model the cap is an ownership lever, not a fertility lever; the July 1 audit claim (TFR +0.071 from the cap) was measured on the earlier package and does not carry over.
- Not done: nothing was fitted, no shock estimation, no transition, no production default changed, per the standing rule. The recovery batch (job 17858740) stays stopped.

4. Baseline and provenance

main equals the tag `paper-baseline-2026-09-14` in the model package and carries the exact solver and the pension helpers; Codex replayed the initial recipe on Torch and reproduced all 13 moments, 17 parameters and the loss 179.298 (`output/model/paper_baseline_sep14/replay_20260917/README.md`). A byte copy of the presented code is under `calibration_archive/legacy_presentation_20260914/`. The sandbox uses the recipe's own assembly functions; two parameter bugs were fixed, and its baseline still differs from the slides' state in two wealth-distribution rows because the recipe starts from a checkpoint that only exists on the cluster. Rule: the cluster replay is the exact tool for levels; the sandbox is the quick tool for differences. A stray git ref `refs/remotes/origin/main` 2 blocks automatic repacking; left for the Codex session.

5. Next steps, in order

1. Done: both Torch runs collected (rows above).
2. Decide F3, P1, H3/H6, H1, E1, F2 from the table; each has a recommendation above.
3. With the decisions made, one refit under the same target contract, on Torch, then the frictionless and constrained diagnostics again on the refit, so the mechanism statement is made once on the model the paper will keep.
4. Rewrite the mechanism paragraph of the slides and the paper around what the model does: liquidity and the price of space move births; the down payment moves ownership and the allocation of large homes.
5. Only then the history refit and the policy transition.

Everything above is also in the follow-up log of `docs/model/structural_model_review_consolidated_20260915.md`, with paths.